

Exciting innovations are at our fingertips, but
what happens to our old gadgets when we
embrace the next trend?



Let's embark on a journey to uncover
the hidden fate of our e-waste.



The Impact of E-Waste

DEFINING E-WASTE

Electronic waste (e-waste) refers to the refuse created by disposed electronic devices, as well as the disposal of any materials involved in their manufacture or use.

E-WASTE: THE FASTEST GROWING MUNICIPAL WASTE STREAM IN AMERICA

The United States produces more e-waste than any other country in the world.

In 2012, the U.S. generated **3.4 MILLION TONS OF E-WASTE**



Americans send approximately **50,000 DUMP TRUCKS WORTH OF ELECTRONICS** to recyclers each year.

E-waste currently represents



E-WASTE ACROSS THE GLOBE

E-WASTE CONTAINS TOXIC COMPONENTS INCLUDING:

- Mercury
- Lead
- Cadmium
- Polybrominated Flame Retardants
- Barium
- Lithium
- Polyvinyl Chloride

IN 2016 **44.7 MILLION METRIC TONS** OF E-WASTE WAS GENERATED. THAT'S THE EQUIVALENT OF

4,500 EIFFEL TOWERS

THE GOLD, SILVER, COPPER, AND OTHER VALUABLE MATERIALS FOUND IN THIS E-WASTE WOULD HAVE BEEN WORTH **\$55 BILLION** HAD THEY BEEN RECOVERED

THINGS YOU DIDN'T KNOW ABOUT

1

Manufacturing of a computer and screen



Resources Weights More than an Rhino.

2

Approximately only 10%-18% of the product is recycled.



3

Energy of the computer

The **ratio** energy of the computer during manufacturing [81%] and operation [19%].



4

Electronics are Valuable

60 EL Elements

Electronic are valuable components, **60 elements** can be found in electronics both **good** and **bad**.

VALUABLE

Gold
Silver
Platinum
Aluminum

HAZARDOUS

Lead, Cadmium
Mercury
Phthalate Plasticizer
Brominate Flame Retardants (BFR)

CATHODE RAY TUBE

PRESENT IN ALL ELECTRONICS
POISONOUS

5

Security Implications

Electronics often contain **confidential information**.

If hard drives were not correctly wiped before they were sold, it **leaves opportunity for fraud**.

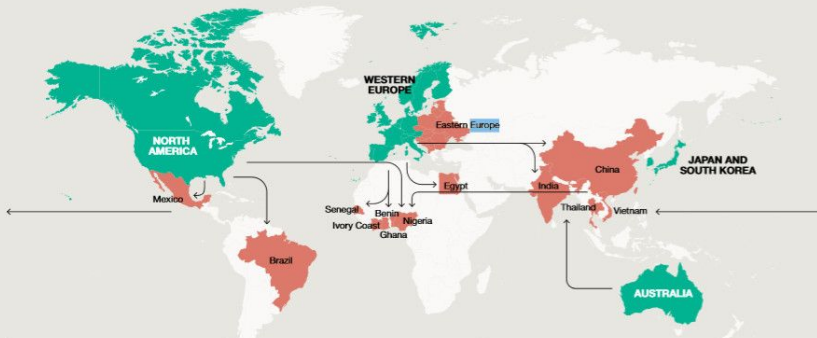


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60 MILLION OF GOLD AND SILVER

Americans dump phones containing **\$60 million in gold and silver** yearly.

MAPPING OUT E-WASTE



Some of the highest and lowest e-waste generating nations: E-Waste generated (\$g per capita), 2018

28.5	24.9	24.8	23.9	23.6	0.4	0.5	0.6	0.6	0.8
Norway	United Kingdom	Denmark	Netherlands	Australia	Niger	Ethiopia	Afghanistan	Uganda	Nepal



TWN

HOW ELECTRONIC WASTE AFFECTS YOUR HEALTH & PLANET

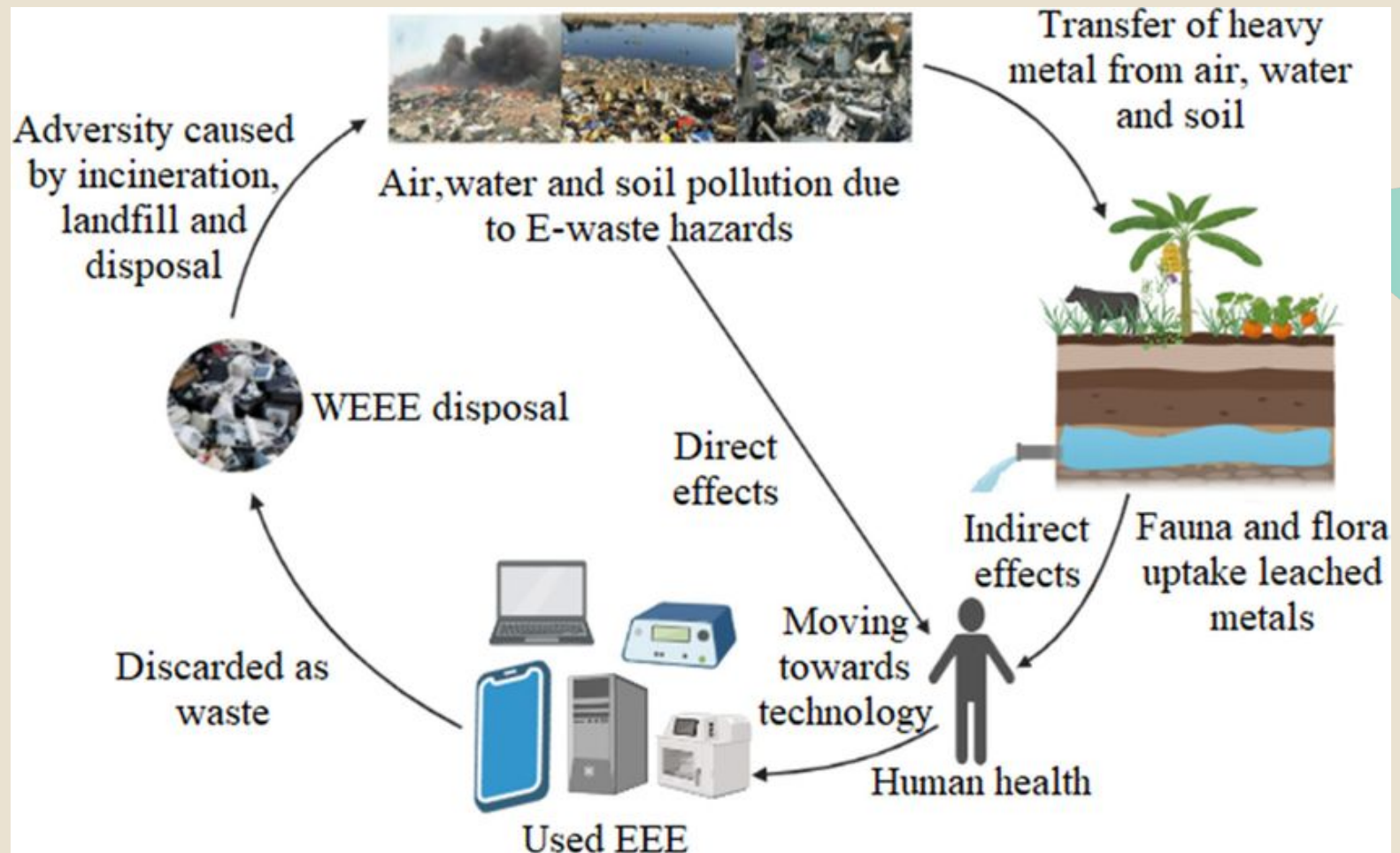


Thousands



Growth of E-waste in India

Weight in MT





Attempts that have been made

Key initiatives:

- E-Waste (Management) Rules, 2016, which introduced Extended Producer Responsibility (EPR)
- Swachh Digital India
- National Clean Energy and Environment Fund



A Holistic Solution to E-Waste Management





E-Waste Collection and Tracking

Smartphone App Integration



Building Partnerships





Blockchain Tracking

IN FACTORY



DISMANTLING



REUSE



PACKED



COLLECT E-WASTE



E-WASTE RECYCLING

14 OCTOBER 2024

International E-Waste Day

Join the e-waste hunt – retrieve, recycle and revive!



E-waste is the fastest growing waste stream in the world

Global e-waste
generated is
projected to reach

74.7 million
tonnes

by 2030

=

7390

Eiffel Towers

Source: Global E-Waste Monitor 2020



International E-Waste Day



E-waste Recycling Campaign

- Adya Srivastava

